Exact Design-Space Exploration for SoC Security Feature Synthesis: A Declarative Formulation with Cross-Solver Validation on OpenTitan

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***Abstract*—Security features in Systems-on-Chip consume the same LUT, FF, power, and latency budget as the functional design, yet security architecture decisions are still made manually. This paper formulates system-level security feature synthesis as an exact multi-objective optimization problem using Answer Set Programming (ASP). Given a component and asset model with impact annotations, the method selects a security mechanism and a monitoring option for every component under explicit FPGA resource and latency constraints, returning provably optimal solutions.**

The formulation is validated on a benchmark suite spanning 1 to 20 components across 27 runs. The largest case is a 20-component instance derived from the open-source OpenTitan Earl Grey root-of-trust SoC, targeting a Xilinx Kintex-7 410T FPGA. To establish correctness independently of the ASP solver, an equivalent Constraint Programming (CP-SAT) formulation is built and cross-validated: all 24 synthetic runs (1–8 components) produce identical proven-optimal objectives across both solvers, establishing formulation correctness. At 20 components, ASP does not converge to the proven optimum within 500 s, while CP-SAT proves optimality in under 0.12 s—a qualitative scalability boundary that motivates a hybrid workflow. The benchmark demonstrates constraint sensitivity at realistic scale: tightening the LUT budget raises optimal risk from 1,149 to 2,139 ($1.86\times$); restricting power forces a mixed security assignment exploiting redundancy-group structure that a greedy heuristic misses by 234% and a local-search refinement still misses by 55%.

***Index Terms*—design-space exploration, security-aware CAD, System-on-Chip, Answer Set Programming, FPGA, feature synthesis, OpenTitan, cross-solver validation**

I. Introduction

Modern SoCs integrate processors, accelerators, memories, and peripherals on a shared interconnect. A security decision at one component—for example, selecting a zero-trust firewall versus a lightweight MAC checker—affects system-wide risk exposure, implementation cost, and timing across the entire device. This makes security architecture a constrained optimization problem: security features, anomaly detectors, and bus monitors improve protection but consume the same LUT, FF, BRAM, and power budget needed by the functional design.

Prior work has argued that security should become a first-class design objective in embedded-system exploration [1], and several papers have explored design-for-security tradeoffs for specific vulnerabilities or application domains [2]–[8]. What is still missing is (i) a clean, implementation-aware formulation for *system-level* feature synthesis solved to proven optimality, (ii) cross-solver validation establishing that the reported optima are correct, and (iii) evaluation on an architecture-realistic benchmark derived from an open-source SoC at a scale that reveals design insights invisible in small synthetic cases.

This paper addresses all three gaps. The security feature synthesis problem is formulated in ASP and solved with clingo [9]. An equivalent CP-SAT formulation using Google OR-Tools [10] is independently constructed and cross-validated on every benchmark instance. The benchmark suite includes eight synthetic cases (1–8 components) and a 20-component instance derived from OpenTitan Earl Grey [11], an open-source silicon root-of-trust SoC with cryptographic accelerators, a key manager, lifecycle controller, and communication peripherals.

The paper makes four contributions:

1) An exact formulation of system-level SoC security feature synthesis under explicit LUT, FF, BRAM, power, and latency constraints, expressed in ASP and solved with clingo.
2) Cross-solver validation: an equivalent CP-SAT model produces identical proven-optimal objectives on all 24 synthetic runs (1–8 components), establishing formulation correctness independently of the solver.
3) A scalability comparison from 1 to 20 components showing that ASP proves optimality up to 8 components (7.2 s mean) but fails at 20, while CP-SAT solves every case in under 0.12 s, motivating a hybrid workflow with CP-SAT as the production feature-synthesis solver.
4) A constraint-sensitivity finding on the 20-component OpenTitan case: tightening the power budget forces a mixed security assignment (9 ZT + 6 DM + 5 MAC) that exploits redundancy-group structure, producing a 234% improvement over greedy and 55% over local search, quantifying the value of exact optimization.

II. Related Work

Security-aware design-space exploration has been studied for over a decade, but most existing work either optimizes for a specific vulnerability or treats security as an auxiliary constraint. Stierand et al. [2] integrated security requirements into DSE but did not calibrate implementation costs. Gressl et al. [5], [6] explored secure IoT alternatives derived from known attacks. Liu and Li [3], Rajmohan et al. [4], and Linares et al. [7] each study narrower implementation spaces around particular protections. Mahfuz et al. [8] use explainable AI for design-for-security exploration, but target rule-guided hardening rather than exact implementation-aware synthesis.

None of these works (i) solve the system-level feature selection problem to proven optimality under FPGA resource constraints, (ii) provide cross-solver validation of the reported optima, or (iii) evaluate on an architecture-realistic benchmark derived from an open-source SoC. This paper fills all three gaps.

On the solver side, ASP has been used for hardware configuration [9] and combinatorial optimization. CP-SAT [10] is a state-of-the-art constraint programming solver. Using both on the same problem provides solver-independent correctness evidence that is standard practice in operations research but rare in security-aware CAD.

III. Problem Formulation

A. System Model

The input is a set of components $\mathcal{C}$, each containing assets with declared read/write impact scores and allowable latency bounds. The synthesis engine assigns one security feature $s \in \{\texttt{mac}, \texttt{dynamic_mac}, \texttt{zero_trust}\}$ and one logging feature $\ell \in \{\texttt{no_logging}, \texttt{some_logging}, \texttt{zero_trust_logger}\}$ to each component. Each feature pair (s, ℓ) has known LUT, FF, BRAM, power, and latency costs drawn from a Vivado-derived catalog, as well as vulnerability (V) and logging (L) scores that determine residual risk.

B. Feature Catalog

Feature costs are derived from Vivado 2023.x synthesis of standalone security IP cores (a MAC checker, a dynamic-MAC engine with nonce management, a zero-trust firewall, and three monitoring/logging engines) targeting a Xilinx xc7z020 (PYNQ-Z2 class) at 125 MHz. Zero-trust resource estimates are based on possible implementations explored in Butka and Bobda [12]. LUT and FF counts are architecture-independent 7-series primitive counts and transfer directly to XC7K410T without correction. Power values are Vivado static-power estimates normalised to 10^{-2} W integers; they are order-of-magnitude indicators, not post-place-and-route measurements. The zero-trust logger's large base cost (31,354 LUTs) reflects a full transaction-logging engine with FIFO and formatter.

Each feature has three cost tiers: a *per-asset* cost (proportional to the number of assets on the component), a *per-component* cost (fixed per protected component), and a *base* cost (one-time overhead, shared across all components using that feature). Table I summarizes the catalog.

Lower V and L scores reduce residual risk (Eq. 1), but stronger features cost more resources. Vulnerability scores are scaled by 10 ($V = 10, 20, 30$) to keep all arithmetic in integers. Zero-trust provides the lowest vulnerability ($V = 10$) but requires more LUTs, FFs, and power per component than MAC or dynamic-MAC (Table I), creating a genuine cost–security tradeoff that the optimizer must resolve under each constraint profile. The zero-trust logger, with $L = 5$, cuts the logging factor $4\times$ relative to no-logging ($L = 20$), but at a base cost of 31,354 LUTs. Whether these tradeoffs are worthwhile depends on the system's impact distribution and resource budget—questions the optimizer answers automatically.

TABLE I
Feature-cost catalog (Vivado synthesis, xc7z020 @ 125 MHz)

Feature	Class	LUTs			FFs			V / L	Pwr[a]
		/asset	/comp	base	/asset	/comp	base		
MAC	sec	51	234	0	32	292	0	$V=30$	2
Dynamic MAC	sec	51	234	1985	32	292	1612	$V=20$	2
Zero Trust	sec	90	380	3200	56	480	2600	$V=10$	3
ZT Logger	log	—	—	31354	—	—	25020	$L=5$	717
Some Logging	log	—	—	3135	—	—	2502	$L=10$	72
No Logging	log	—	—	0	—	—	0	$L=20$	0

Base costs are shared across components that use the same feature.
[a] Power (10^{-2} W): per-component for security, one-time base for logging.

C. Risk Objective

For each component c, asset a, and operation op, residual risk follows the multiplicative model of Butka and Bobda [12]:

$$r(c, a, op) = \left\lfloor \frac{I(a, op) \cdot V(s_c) \cdot L(\ell_c)}{10} \right\rfloor \tag{1}$$

where I is the declared impact (1–10), V is the vulnerability score of the selected security feature s_c ($V = 10, 20, 30$; lower is better), and L is the logging score ($L = 5, 10, 20$; lower is better). Both V and L are scaled by 10 for integer arithmetic; the divisor compensates, giving effective risk multipliers in $[5, 600]$.

The multiplicative form follows the standard risk = impact $\times$ likelihood decomposition (ISO 27005, NIST SP 800-30): V quantifies residual exposure after the selected prevention mechanism, and L quantifies residual observability gap (a component with full logging has lower undetected-compromise likelihood). The product $V \cdot L$ is thus a proxy for residual likelihood given the chosen feature pair, and multiplication with I yields a composite risk score. We stress that this is an *engineering scoring heuristic*, not a calibrated probability; V and L are interval-scaled design parameters chosen to be proportional to residual exposure, not ordinal rankings. A per-asset risk cap $r(c, a, op) \leq R_{\max}$ ensures no single asset exceeds an acceptable threshold.

The global objective minimizes total system risk:

$$\min \sum_{c,a,op} r(c, a, op) \tag{2}$$

subject to per-resource constraints (LUTs, FFs, BRAM, power, latency) and the per-asset risk cap.

D. OpenTitan Earl Grey Benchmark

To move beyond synthetic architectures, we derive a 20-component benchmark from the OpenTitan Earl Grey [11] architecture, an open-source silicon root-of-trust SoC designed by lowRISC. The component set (Table II) maps one-to-one with Earl Grey IP blocks; impact scores are assigned by the paper authors to reflect each block's architectural criticality (not measured vulnerability data): cryptographic engines and the key manager receive the highest scores; communication

TABLE II
OPENTITAN EARL GREY COMPONENT MAPPING (20 IPS)

Comp.	OpenTitan IP	I_r	I_w	Role
cpu	Ibex RISC-V	10	10	Master processor
dma	DMA controller	8	8	Bus master
aes	AES accelerator	9	10	Symmetric crypto
hmac	HMAC	8	9	Authentication
kmac	KMAC	8	9	Keccak MAC
otbn	OTBN bignum	9	10	Asymmetric crypto
keymgr	Key manager	10	10	Root of trust
otp	OTP controller	9	9	Immutable fuses
lc	Lifecycle ctrl	9	8	State transitions
flash	Flash ctrl	7	8	Code/data storage
sram	SRAM ctrl	5	6	Working memory
rom	ROM ctrl	6	3	Boot code
uart0	UART 0	3	3	Serial I/O
uart1	UART 1	3	3	Serial I/O
gpio	GPIO	2	4	General I/O
spi	SPI host	4	5	SPI interface
i2c	I2C	3	4	I2C interface
timer	Timer	2	2	Non-critical
alert	Alert handler	7	5	Security monitor
entropy	Entropy source	8	7	RNG + health

peripherals the lowest. Latency caps follow the same logic: crypto blocks have tight bounds (5–10 cycles), peripherals are relaxed (100–1000). The benchmark is therefore "architecture-derived"—it preserves OpenTitan's component topology and role differentiation—but relies on the external security-IP cost catalog (Table I), not on OpenTitan's actual gate-level implementations.

Two redundancy groups model defense-in-depth coverage: AES/HMAC/KMAC (crypto accelerators that jointly protect key material—compromise of one does not expose data protected by the others) and UART0/UART1 (redundant communication channels). Groups are not claims of functional interchangeability; they assert that the members provide overlapping security coverage so that combined residual exposure is lower than any single member's.

E. Redundancy Model

For components not in any group, risk follows Eq. 1 directly. For components in a redundancy group $G = \{c_1, \ldots, c_n\}$, the formulation combines per-member residual probabilities before computing risk, modelling independent exposure reduction.

Each member's $V \cdot L$ product is normalised to a common scale using parameters $\mu = 25$ (floor) and $\omega = 1000$ (ceiling):

$$p_{\text{norm}}(c) = \frac{(V_c \cdot L_c - \mu) \cdot 1000}{\omega - \mu} \tag{3}$$

The combined normalised probability is the recursive product:

$$P_G = \frac{p_{\text{norm}}(c_1) \cdot p_{\text{norm}}(c_2) \cdots p_{\text{norm}}(c_n)}{1000^{\,n-1}} \tag{4}$$

which is then denormalised back: $p_{\text{denorm}} = P_G \cdot (\omega - \mu)/1000 + \mu \cdot 10$. Group-member risk becomes $r(c, a, op) = \lfloor I(a, op) \cdot p_{\text{denorm}}/100 \rfloor$. The 1000^{n-1} divisor ensures that adding a redundant member always reduces combined risk, giving the optimizer an incentive to accept weaker per-member security when the group already provides coverage.

TABLE III
BENCHMARK SUITE

Case	Comp.	Assets	Groups	Target
TC1	3	3	no	Synthetic
TC2	8	8	no	Synthetic
TC3	1	8	no	Synthetic
TC5	8	8	no	Synthetic
TC6	3	3	yes	Synthetic
TC7	8	8	yes	Synthetic
TC8	8	8	yes	Synthetic
TC9	8	8	yes	Synthetic
OT	20	20	yes	OpenTitan Earl Grey

IV. SOLVER FORMULATIONS

A. ASP Formulation

The problem is encoded in ASP and solved with clingo 5.8.0 [9]. Each component receives exactly one security and one logging feature:

$$\sum_{s \in F_{\text{sec}}} \text{sel_sec}(c, s) = 1, \quad \sum_{\ell \in F_{\text{log}}} \text{sel_log}(c, \ell) = 1 \tag{5}$$

Hard constraints enforce device budgets:

$$\text{LUTs}_{\text{used}} \leq \text{LUTs}_{\max}, \quad \text{Power}_{\text{used}} \leq \text{Power}_{\max} \tag{6}$$

$$\forall (c, a, op): \ \text{latency}(c, a, op) \leq \text{cap}(a, op) \tag{7}$$

The objective minimizes total residual risk under these constraints. The ASP encoding is declarative: adding a new security feature requires only adding catalog facts, not modifying the solver logic. This extensibility is the primary motivation for choosing ASP over imperative formulations.

B. CP-SAT Formulation

To validate the ASP results independently, an equivalent model is built using Google OR-Tools CP-SAT [10]. The CP-SAT model uses:

- **Element constraints** to look up per-pair (security × logging) resource costs and $V \cdot L$ products from precomputed tables.
- **Multiplication and division equalities** to compute $I \cdot V \cdot L/10$ for each asset.
- **Boolean indicators** with conditional constraints to track one-time base costs for each feature used.

Both formulations encode the identical optimization problem. Any objective mismatch would indicate a bug in one or both encodings.

V. EXPERIMENTAL SETUP

A. Benchmark Suite

The suite contains nine instance families: eight synthetic cases (TC1–TC9, excluding TC4, which uses a hierarchical component model with a different arity than the flat `component/1` encoding shared by the remaining cases and both solvers) and the OpenTitan-derived case (OT). Table III summarizes the benchmark.

TABLE IV
CONSTRAINT PROFILES

Profile	Target	LUT cap	Power cap	Risk cap
A	xc7z020	53,200	15,000	100
B	xc7z020	10,000	15,000	500
C	xc7z020	53,200	1,000	100
OT-A	XC7K410T	254,200	15,000	100
OT-B	XC7K410T	25,000	15,000	500
OT-C	XC7K410T	254,200	800	100

B. Constraint Profiles

Each synthetic case is evaluated under three profiles (Table IV) targeting a Xilinx xc7z020 (PYNQ-Z2). The OpenTitan case uses three profiles targeting a Xilinx Kintex-7 410T, matching the FPGA used by OpenTitan implementations. OT-A provides full resources; OT-B restricts LUTs to 25,000 (below the 43,954 required by the OT-A optimum); OT-C restricts power to 800 (below the 820 required by OT-A), forcing the optimizer to substitute cheaper security features on some components.

C. Experimental Environment

Experiments ran on one P-core of an Intel Core i7-12700H (2.3 GHz) with 32 GB RAM, Windows 11, Python 3.12, clingo 5.8.0, and OR-Tools 9.x. Both solvers were run single-threaded for fair comparison.

VI. RESULTS

A. Cross-Solver Validation

Table V reports the complete benchmark results. All 24 synthetic runs (1–8 components) produce identical proven-optimal objectives between ASP and CP-SAT, establishing that both formulations encode the same optimization problem. At 20 components (OpenTitan), CP-SAT proves optimality in under 0.12 s, but ASP does not converge to the proven optimum within a 120 s timeout—it finds feasible but suboptimal solutions after 350–520 s of search, demonstrating a qualitative scalability boundary between 8 and 20 components.

B. Scalability Analysis

Table VI aggregates solve time by component count. At 1 component, both solvers complete in under 5 ms. At 3 components, ASP takes 0.16 s (dominated by grounding), while CP-SAT stays under 15 ms. At 8 components, ASP averages 7.2 s due to the wide optimization range created by the $10\times$-scaled vulnerability scores, while CP-SAT remains under 30 ms—a $362\times$ speedup. At 20 components (OpenTitan), ASP does not converge to the proven optimum within 500 s, while CP-SAT proves optimality in under 0.12 s. The scalability boundary between 8 and 20 components marks where CP-SAT becomes essential for practical use.

Figure 1 plots the scaling trend on a log scale. The gap between solvers widens steadily: $1.5\times$ at 1 component, $14\times$ at 3, $362\times$ at 8, and $>5{,}600\times$ at 20 where ASP fails to prove optimality entirely. For rapid parameter sweeps or larger designs, CP-SAT is the pragmatic feature-synthesis choice; ASP's value lies in the declarative extensibility required by downstream reasoning stages (Section VII-A).

TABLE V
CROSS-SOLVER VALIDATION: ALL 27 RUNS

Case	Comp.	Prof.	Obj.	ASP (s)	CP-SAT (s)	Match
TC1	3	A	90	0.17	0.012	✓
TC1	3	B	180	0.15	0.005	✓
TC1	3	C	90	0.18	0.007	✓
TC2	8	A	240	7.21	0.013	✓
TC2	8	B	540	8.13	0.015	✓
TC2	8	C	240	7.04	0.017	✓
TC3	1	A	240	0.004	0.003	✓
TC3	1	B	480	0.004	0.003	✓
TC3	1	C	240	0.005	0.002	✓
TC5	8	A	240	6.99	0.014	✓
TC5	8	B	540	7.53	0.014	✓
TC5	8	C	240	6.50	0.013	✓
TC6	3	A	45	0.16	0.015	✓
TC6	3	B	45	0.15	0.013	✓
TC6	3	C	45	0.15	0.015	✓
TC7	8	A	172	7.21	0.025	✓
TC7	8	B	282	6.77	0.025	✓
TC7	8	C	172	7.96	0.028	✓
TC8	8	A	206	6.71	0.018	✓
TC8	8	B	383	6.95	0.017	✓
TC8	8	C	206	6.83	0.018	✓
TC9	8	A	159	7.60	0.029	✓
TC9	8	B	244	6.74	0.027	✓
TC9	8	C	159	7.48	0.026	✓
OT	20	OT-A	1,149	$>347^{\dagger}$	0.11	—
OT	20	OT-B	2,139	$>382^{\dagger}$	0.04	—
OT	20	OT-C	1,379	—	0.06	—

†ASP suboptimal; objectives are CP-SAT proven optima.

TABLE VI
SCALABILITY: MEAN SOLVE TIME BY COMPONENT COUNT

Components	ASP mean (s)	CP-SAT mean (s)	Ratio
1	0.004	0.003	$1.5\times$
3	0.16	0.011	$14\times$
8	7.18	0.020	$362\times$
20	$>400^{\dagger}$	0.071	$>5{,}600\times$

†ASP did not prove optimality; best-found was suboptimal.

C. OpenTitan Design Insights

Table VII reports the OpenTitan results in detail.

Finding 1: Resource constraints raise optimal risk and force heterogeneous assignments. Under Profile OT-A (full resources), the optimizer achieves an optimal risk of 1,149 using 43,954 LUTs and 820 power units, with zero-trust security and zero-trust logging on all 20 components. When LUTs are capped at 25,000 (OT-B), the optimizer cannot afford

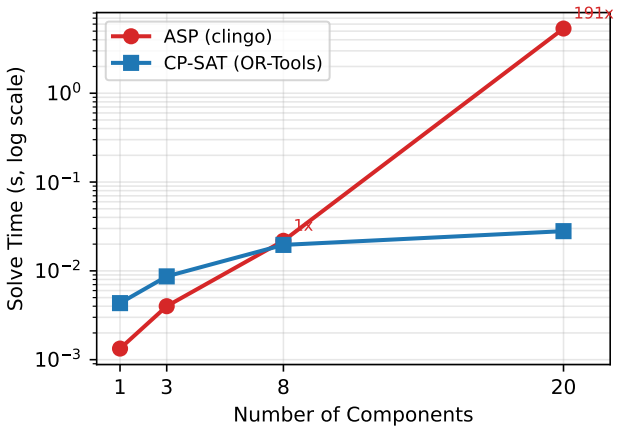

Fig. 1. Mean solve time vs. component count (log scale). The gap widens from 1.5× at 1 component to >5,600× at 20 components, where ASP cannot prove optimality.

TABLE VII
OPENTITAN EARL GREY RESULTS (20 COMPONENTS, CP-SAT PROVEN OPTIMA)

Profile	Obj.	LUTs	Power	Security mix	Logging mix
OT-A	1,149	43,954	820	20 ZT	20 ZTL
OT-B	2,139	15,735	175	20 ZT	20 SL
OT-C	1,379	43,904	800	9 ZT+6 DM+5 MAC	20 ZTL

ZT=Zero Trust, DM=Dynamic MAC, MAC=MAC, ZTL=ZT Logger, SL=Some Logging

the ZTL base cost (31,354 LUTs) and switches all logging to some-logging, raising risk to 2,139 (1.86×). When power is capped at 800 (OT-C), the optimizer maintains full ZTL logging but downgrades security on 11 of 20 components: 6 receive dynamic-MAC and 5 receive MAC, raising risk to 1,379 (1.20×).

Finding 2: The optimizer exploits cost differentiation and redundancy. Under OT-C, the optimizer assigns ZT to the nine highest-criticality components and cheaper features to the rest. The three crypto accelerators (aes, hmac, kmac) receive MAC despite high individual impact—their redundancy group membership reduces combined probability, making ZT's risk saving less valuable than MAC's power saving. This redundancy-aware assignment requires exact optimization.

Finding 3: The first casualty depends on the binding constraint. OT-B (LUT-constrained) downgrades logging while keeping full ZT security; OT-C (power-constrained) downgrades security while keeping full ZTL logging. Which dimension degrades first depends on which resource binds, not on a fixed priority.

D. Heuristic Baseline Comparison

To quantify the benefit of exact optimization, Table VIII compares CP-SAT optima against two heuristic baselines: a *greedy* heuristic that processes components by descending impact and assigns the best affordable feature, and a *local-search* (1-opt) refinement that starts from the greedy solution and iteratively swaps one component's security/logging assignment to the best feasible improving neighbour until no single-component swap reduces risk.

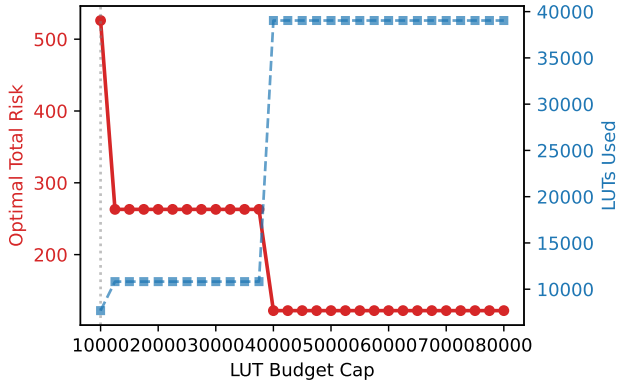

Fig. 2. Pareto sweep: optimal risk vs. LUT budget cap on the OpenTitan case. Sharp transitions correspond to logging-tier and security-tier thresholds; mixed security assignments appear under tight budgets.

TABLE VIII
HEURISTIC VS. OPTIMAL (CP-SAT) ON OPENTITAN PROFILES

Profile	Optimal	Greedy	Gap_G	Local-S	Gap_{LS}
OT-A	1,149	1,149	0%	1,149	0%
OT-B	2,139	2,139	0%	2,139	0%
OT-C	1,379	4,599	+234%	2,139	+55%

When constraints permit a uniform solution (OT-A, OT-B), both heuristics match the optimum. Under OT-C's power constraint, greedy commits ZT to high-impact components first, exhausting the budget and forcing late components into MAC with no-logging (3.3× worse risk). Local search reduces the gap from 234% to 55% but cannot coordinate the multi-component trades needed to exploit redundancy. The residual 55% gap confirms that exact optimization matters when resources force heterogeneous assignments.

E. Perturbation Stability

All 24 synthetic runs produce identical proven optima across ASP and CP-SAT, validating the formulation on 1–8 component instances. Profile C always reproduces Profile A at 8 components (power is never binding), whereas on OpenTitan, OT-C produces a dramatically different solution with mixed security assignments—power sensitivity and cost differentiation emerge only at realistic scale.

VII. DISCUSSION

A. Why ASP for Security Feature Synthesis

At 8 components CP-SAT is 362× faster than ASP; at 20 components ASP cannot prove optimality at all, while CP-SAT completes in under 0.12 s (Table VI). For feature-synthesis alone, CP-SAT is the production solver. The case for ASP rests on the downstream reasoning stages. Feature synthesis is the first of three stages; the maintained repository also implements *access-control policy synthesis* (firewall

placement, least-privilege gap analysis via negation-as-failure) and *resilience analysis* (parallel transitive closures over degraded topologies, bounded-depth attack-path enumeration). Both stages rely on recursive reachability, negation-as-failure, and multi-level implication chains—patterns natural in ASP but costly in CP-SAT or ILP. A hybrid workflow is therefore attractive: CP-SAT for feature-synthesis optimization, ASP for the downstream stages.

B. Relation to ILP/MIP Solvers

The resource constraints are linear and the objective can be linearised, so ILP solvers (e.g. Gurobi [13]) could solve the same problem. We chose CP-SAT because it is open-source and its element constraints map naturally to feature selection. An ILP comparison remains future work.

C. Constraint Sensitivity as a Design Tool

The OT results demonstrate a practical application of exact DSE: quantifying the security cost of resource trade-offs. A system designer who must fit the security architecture into 25,000 LUTs (instead of the 43,954 required by the unconstrained optimum) now knows the precise cost: risk rises from 1,149 to 2,139 ($1.86\times$), achieved by substituting zero-trust logging with some-logging. Restricting power to 800 units forces a mixed security assignment (9 ZT + 6 DM + 5 MAC) that exploits redundancy-group structure to minimise risk at 1,379 ($1.20\times$)—a non-obvious assignment that the greedy heuristic misses by 234% and local search still misses by 55%. The Pareto sweep (Figure 2) provides a complete risk–resource tradeoff map. This kind of quantitative what-if analysis is only possible with an exact solver that can report provably optimal solutions under each constraint set.

D. Limitations

The OpenTitan benchmark uses impact scores assigned by the paper authors based on architectural criticality, not measured vulnerability data. The feature catalog is derived from Vivado synthesis of standalone security IP cores (MAC checker, dynamic-MAC engine, zero-trust firewall, and three logging monitors), not the actual OpenTitan blocks. The formulation addresses only the feature-synthesis stage; the policy and resilience stages described above are implemented in the maintained repository but are not claimed as contributions of this paper. The benchmark family tops out at 20 components; scaling to 50–100 component SoCs remains future work.

VIII. Conclusion

This paper presented an exact ASP formulation of system-level SoC security feature synthesis, cross-validated against an independent CP-SAT model on 27 benchmark runs spanning 1 to 20 components. All 24 synthetic runs (1–8 components) produce identical proven-optimal objectives, establishing solver-independent correctness. At 20 components, CP-SAT proves optimality in under 0.12 s while ASP does not converge within 500 s, establishing a clear scalability boundary. The 20-component OpenTitan Earl Grey case demonstrates that resource pressure forces non-obvious mixed security assignments exploiting redundancy-group structure—a result that greedy misses by 234% and local search by 55%.

The cross-solver comparison motivates a hybrid workflow: CP-SAT as the production feature-synthesis solver, ASP for the downstream policy-synthesis and resilience stages whose recursive reachability, negation-as-failure, and multi-level implication patterns are natural in ASP but costly in imperative solvers.

Future work will co-optimize firewall and policy placement with feature selection, evaluate resilience under component-compromise scenarios on the OpenTitan topology, and add an ILP solver as a third comparison point.

The full ASP solver, CP-SAT benchmark, feature catalog, and all benchmark instances are available in the accompanying repository.[1]

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